POLITECNICO MILANO 1863

BASI DI DATI 1 – Anno Accademico 2024-2025

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Appello 11/06/2025, **DURATA: 2 ore**

Part 1

Exercise 1 – SQL (13pt)

Consider the following diagram, relating to a bicycle reservation and rental system, where the time stamps can assume full (8 a.m. to 7 p.m.).

CUSTOMER (Num, Name, IDType, IDNumber)
BOOK (Cust-Num, BikeType, Day, StartTime, EndTime)
USE (Cust-Num, Bike-Num, Day, StartTime, EndTime, Cost)
BIKE (Bike-Num, BikeType, CostPerHour, CostPerDay)

A. Write SQL commands to create the BOOK and BIKE tables, making appropriate reasonable assumptions about domains and constraints. (2 pts)
B. List the name of customers who used a bike of a different type than the one they reserved *[note: a specific "use" is relative to a specific "booking" when it occurs on the same day and included time slot]* (3 pts)
C. List the name of customers who have reserved a bike more than once, and who have never used any of them (4 pts)
D. For each customer, indicate the bike used for the maximum amount of hours and the corresponding total number of hours (4 pts)

Exercise 2 – Formal languages (6pt)

Express in Optimized Relational Algebra (3pt) and Datalog (3pt) the following query:

Find the name of the customers who reserved bikes of type "racing" without ever using racing bikes.

PART 2

Exercise 3 – Theory question (1pt):

Describe what is the role of relationship normalization within data design.

Exercise 4 – Conceptual and logical design (13pt)

- *The project evaluation is based on its completeness, correctness, legibility, minimality and clarity. Remember to specify the **identifiers** of all the entities and the **cardinality** of all the associations.*
- *Arrange the conceptual project and the logical project on the inside pages of the sheet.*

An Italian firm provides "gifts" of prepaid experiences in the form of purchase vouchers that can be used at a network of participating supplier. Each purchasable "gift" has a unique internal code assigned by the firm, a predefined cost in Euros, and belongs to a particular category: we consider "typical restaurants," "spas," and "wine bars." For each of these categories, the firm arranges specific suppliers of the service (e.g., restaurant, spa, etc.) in the catalog. Each supplier has an internal code, vat number, name, city, and address, and may propose multiple "gift" alternatives differentiated by commercial description (e.g., "typical Sardinian menu" or "typical fish menu"), each with a fixed unit cost, known only to the firm offering the "gifts" for sale and not visible to the customer.

Buyers of the "gifts" are individuals, characterized by social security number, name, e-mail, mobile contact. The buyer establishes the names of those who will use the gift (can be a person or a couple of people, characterized by a single name).

To take advantage of the gift, the recipient chooses a supplier and goes there on a certain date, after making a reservation in which he or she indicates the internal code of the gift; the system must record the gifts that the customer has actually taken advantage of so that the company can pay the supplier within two weeks from the date of use.